ESM_project

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1 Original Path model

This it the original path model to be tested :

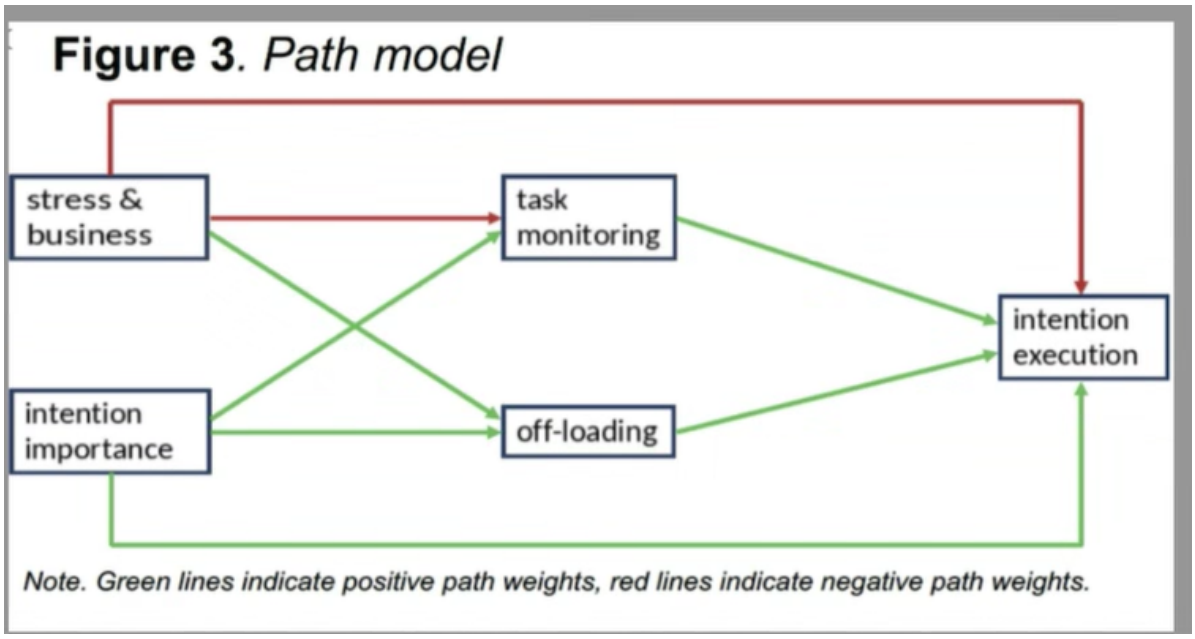


Figure 1: Figure 1. Hypothesised path model

2 Extended model

This model can be extended to accommodate partition of variance between- and within- participants:

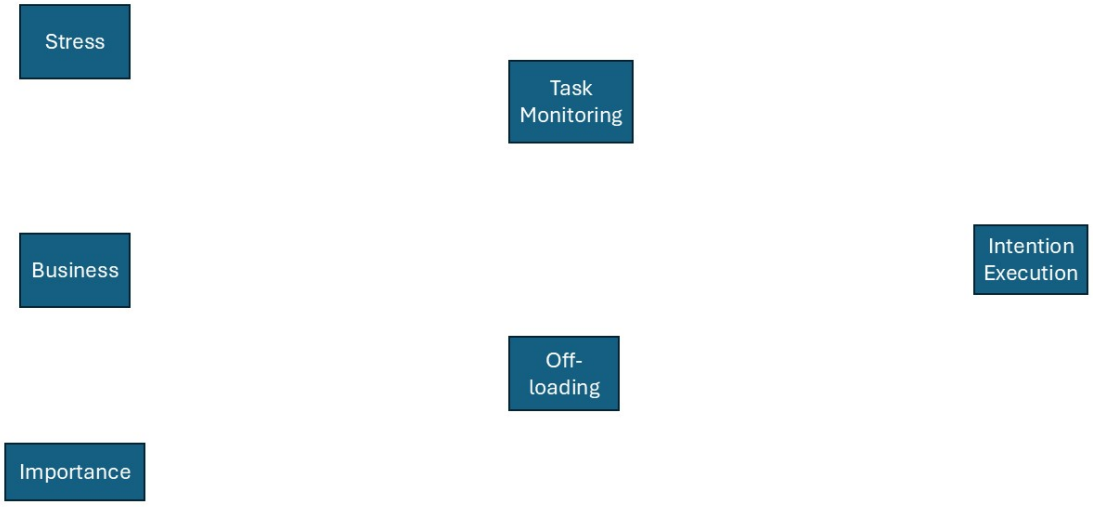


Figure 2: Extended hypothesised path model

3 Within-between partitioning

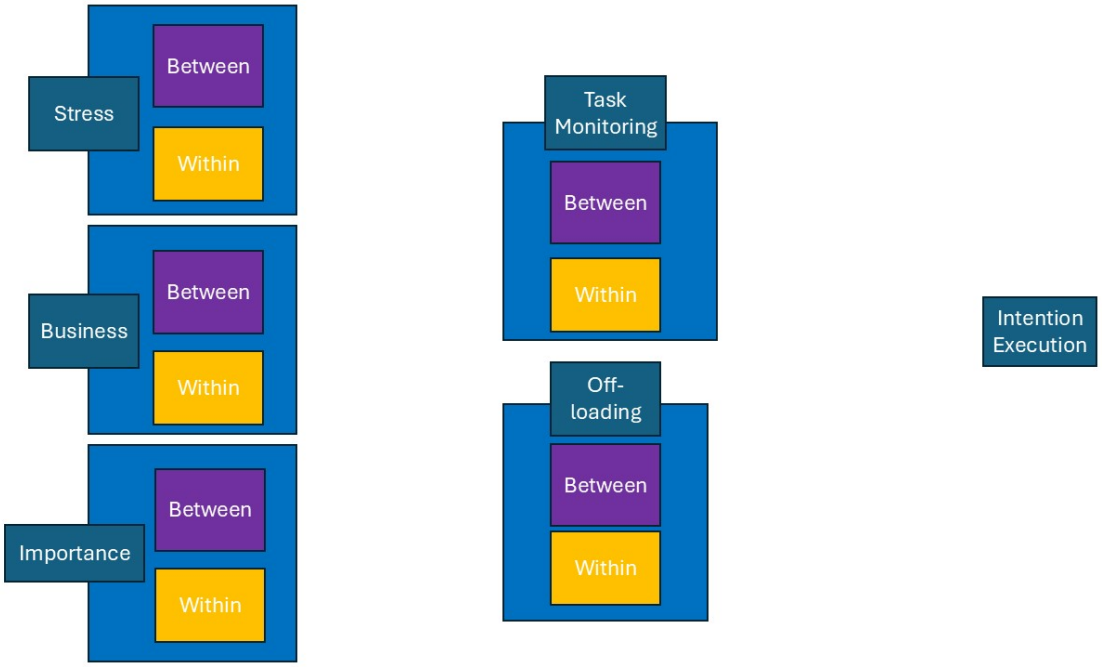


Figure 3: Extended hypothesised path model

4 Effect of stress

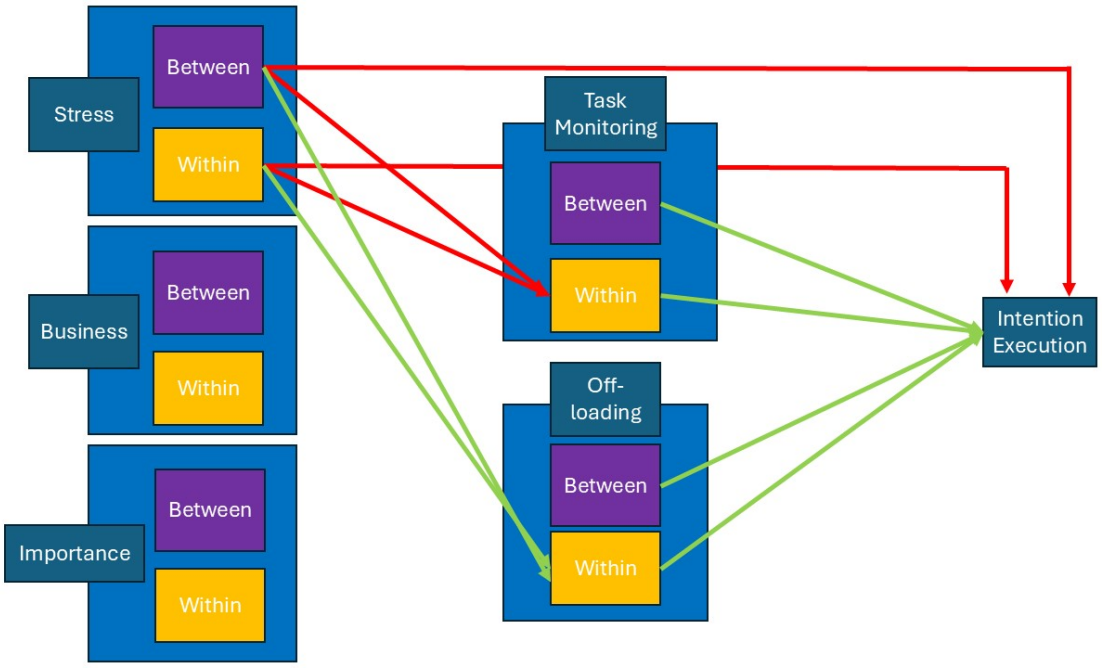


Figure 4: Extended hypothesised path model

5 Effect of business

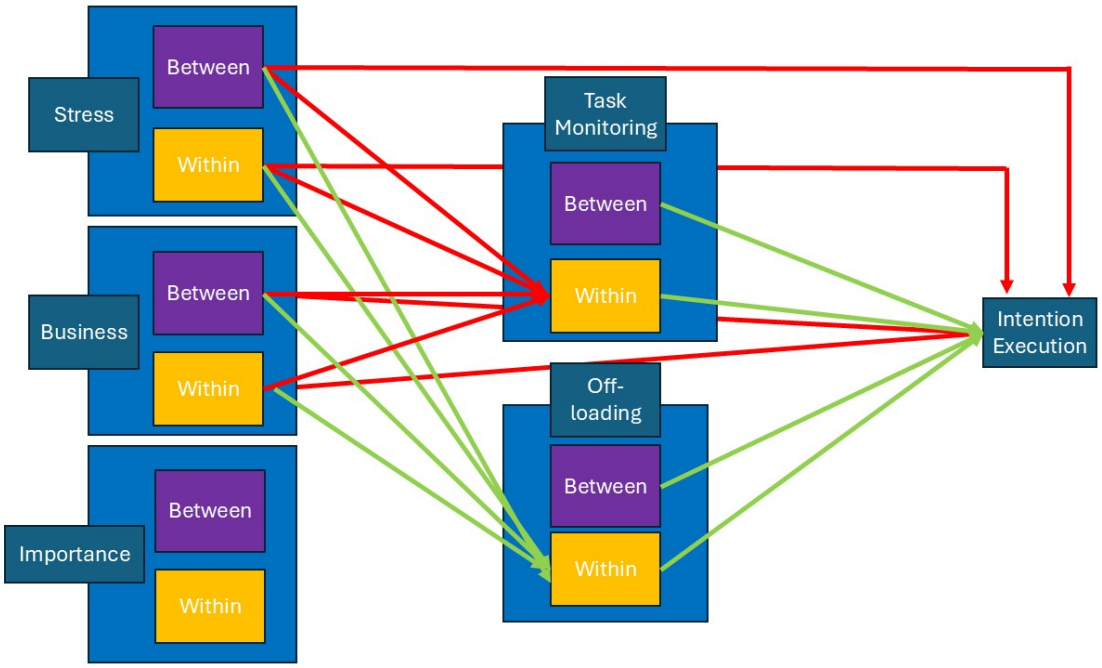


Figure 5: Extended hypothesised path model

6 Effect of Importance

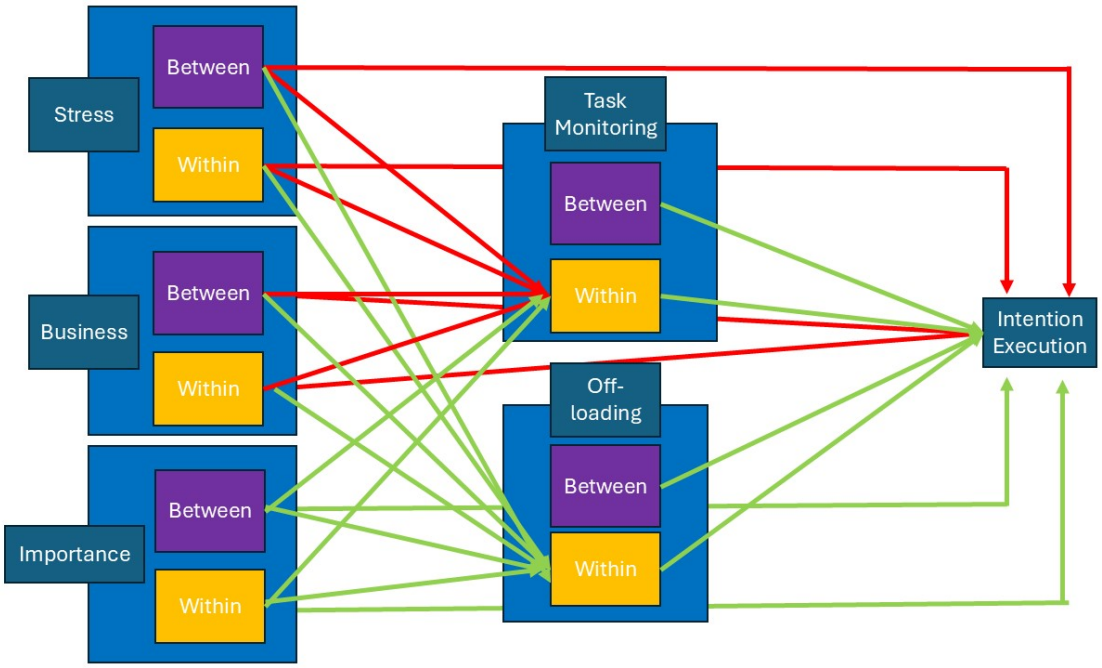
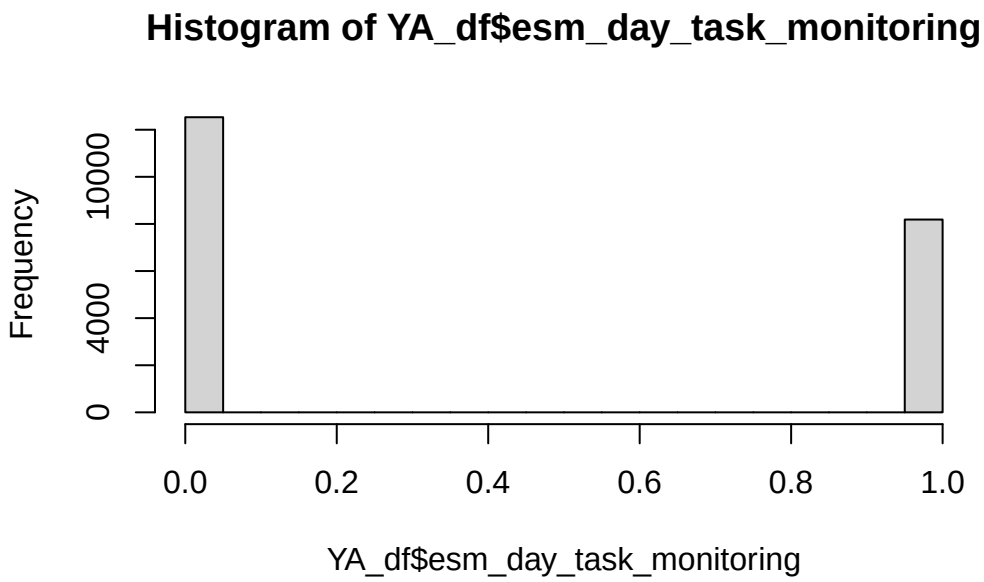


Figure 6: Extended hypothesised path model

```
# Import the data
#| include: false
#| message: false
#| warning: false
#| echo: false
cd<-getwd()

# file dir
file_dir<-"Originaldatensätze/R-Skript"

# young adults' dataset
YA_df<- read_spss(paste0(file_dir, "/ExpSamp_jungErw2.sav"))

# rename the age group
YA_df$agegroup<-"YA"

# check task_monitoring
hist(YA_df$esm_day_task_monitoring)
```




```
# middle-aged adults
MAA_df<-read_spss(paste0(file_dir, "/ExpSamp_mittlErw.sav"))

# rename the age group
MAA_df$agegroup<-"MAA"

# check task_monitoring
hist(YA_df$esm_day_task_monitoring)

# combine the two
all_df<-rbind(YA_df, MAA_df)
```

7 Participants

```
all_df %>%
  distinct(ID, agegroup) %>%
  count(agegroup, name = "n_participants") %>%
  gt() %>%
  tab_header(
```

[

Participants by age group] Participants by age group

agegroup	n_participants
MAA	103
YA	107

```
    title = "Participants by age group"
  )
```

```
# check the length of unique IDs
length(unique(all_df$ID))
```

```
[1] 210
```

8

the df looks like this:

```
# A tibble: 20 x 15
   ID         day  time intention                esm_day_stress esm_day_business
   <chr>    <dbl> <dbl> <chr>                            <dbl>            <dbl>
 1 AM05UN20     1     1 Brief abgeben                        3                2
 2 AM05UN20     1     2 Brief abgeben                        4                3
 3 AM05UN20     1     3 Brief abgeben                        4                5
 4 AM05UN20     1     4 Brief abgeben                        4                4
 5 AM05UN20     1     5 Brief abgeben                       NA               NA
 6 AM05UN20     1     1 Präsentation fertigstel~            3                2
 7 AM05UN20     1     2 Präsentation fertigstel~            4                3
 8 AM05UN20     1     3 Präsentation fertigstel~            4                5
 9 AM05UN20     1     4 Präsentation fertigstel~            4                4
10 AM05UN20     1     5 Präsentation fertigstel~           NA               NA
11 AM05UN20     1     1 Geschwister treffen                 3                2
12 AM05UN20     1     2 Geschwister treffen                 4                3
13 AM05UN20     1     3 Geschwister treffen                 4                5
14 AM05UN20     1     4 Geschwister treffen                 4                4
15 AM05UN20     1     5 Geschwister treffen                NA               NA
16 AM05UN20     2     1 Kaffe trinken gehen                 2                2
```

```
17 AM05UN20     2     2 Kaffe trinken gehen                    2                3
18 AM05UN20     2     3 Kaffe trinken gehen                    4                4
19 AM05UN20     2     4 Kaffe trinken gehen                    5                5
20 AM05UN20     2     5 Kaffe trinken gehen                    5                5
# i 9 more variables: esm_day_task_monitoring <dbl>, esm_day_stress_avg <dbl>,
#   esm_daybusiness_avg <dbl>, esm_day_task_monitoring_avg <dbl>,
#   esm_evening_intention_importance_t1 <dbl>, esm_evening_off_loading <dbl>,
#   esm_evening_intention_execution <dbl>, age_group <dbl>, agegroup <chr>
```

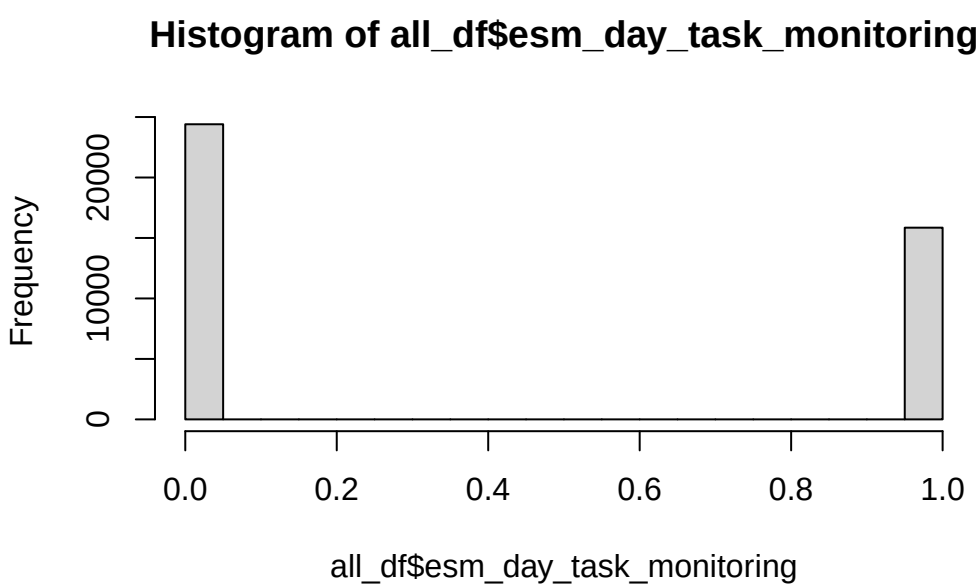
9 Preprocessing

- Rescale PM to range between 0 and 1

```
all_df$esm_evening_intention_execution<-ifelse(all_df$esm_evening_intention_execution ==-1, (
```

10 Monitoring

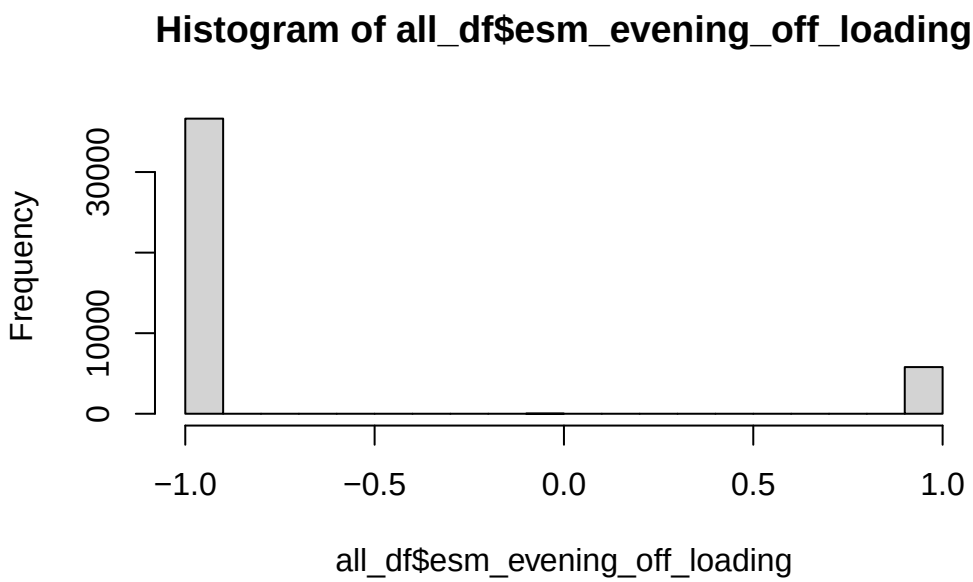
```
hist(all_df$esm_day_task_monitoring)
```



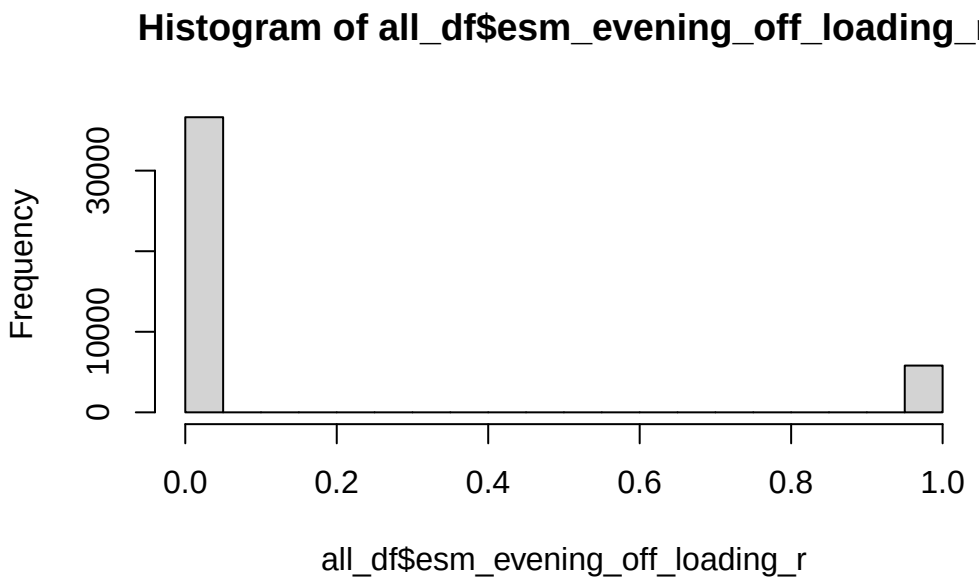
```
# all_df$esm_day_task_monitoring_r<-ifelse(all_df$esm_day_task_monitoring ==-1, 0,
#                                          ifelse(all_df$esm_day_task_monitoring== 0, 0, 1))
```

11 Rescale offloading

```
hist(all_df$esm_evening_off_loading)
```




```
all_df$esm_evening_off_loading_r<-ifelse(all_df$esm_evening_off_loading ==-1, 0,1)

hist(all_df$esm_evening_off_loading_r)
```



12 Reshaping

- The dataset as it is now is good to test hypotheses on the variables that vary within day, e.g. stress, business, monitoring.
- Reshape the dataset so there is only one unique intention per day, which is our base level.

RTeshaped dataset

```
`summarise()` has grouped output by 'ID'. You can override using the `.groups`
argument.
```

```
# A tibble: 20 x 43
# Groups:   ID, day, intention [20]
   ID         day  time intention                esm_day_stress esm_day_business
   <chr>    <dbl> <dbl> <chr>                             <dbl>            <dbl>
 1 AM05UN20     1     1 Brief abgeben                         3                2
 2 AM05UN20     1     3 Geschwister treffen                   4                5
 3 AM05UN20     1     3 Präsentation fertigstel~              4                5
 4 AM05UN20     2     1 Freund besuchen                       2                2
 5 AM05UN20     2     5 Kaffe trinken gehen                   5                5
```

```
 6 AM05UN20     2     4 telefonieren                          5                5
 7 AM05UN20     3     1 Präsentation fertigstel~              2                3
 8 AM05UN20     3     1 aufräumen                             2                3
 9 AM05UN20     3     4 mit Freundin spielen                  4                5
10 AM05UN20     4     3 13 Uhr zum Geburtstag g~              4                5
11 AM05UN20     4     3 mit guter Laune aufsteh~              4                5
12 AM05UN20     4     4 nicht einschüchtern las~              4                5
13 AM05UN20     5     2 18 Uhr bereitmachen für~              3                4
14 AM05UN20     5     3 Motivation zeigen                     4                5
15 AM05UN20     5     1 ausschlafen                           4                4
16 AM05UN20     6     1 Praktikumsplatz suchen                2                4
17 AM05UN20     6     3 Präsentation fertigstel~              4                4
18 AM05UN20     6     3 mit Freunden treffen                  4                4
19 AM05UN20     7     1 Präsentation fertigstel~              5                5
20 AM05UN20     7     5 meinem Freund eine Freu~              2                3
# i 37 more variables: esm_day_task_monitoring <dbl>, esm_day_stress_avg <dbl>,
#   esm_daybusiness_avg <dbl>, esm_day_task_monitoring_avg <dbl>,
#   esm_evening_intention_importance_t1 <dbl>, esm_evening_off_loading <dbl>,
#   esm_evening_intention_execution <dbl>, age_group <dbl>, agegroup <chr>,
#   esm_evening_off_loading_r <dbl>, mean_stress <dbl>, mean_business <dbl>,
#   grand_mean_esm_stress <dbl>, grand_sd_esm_stress <dbl>,
#   grand_mean_esm_day_business <dbl>, grand_sd_esm_day_business <dbl>, ...
```

13 Some particiants have more or less intention per day

```
[1] "Number of unique intentions per day"
```

```
[1] 3 4 2 1
```

```
[1] "Participants who reported 4 intentions in some days:"
```

```
 [1] "CH09LD19" "EK13AN09" "EL05AN19" "EL08TZ04" "EL09ST09" "ER06AH19"
 [7] "ER06SA06" "ER08IG13" "ER09AN17" "ER11RD06" "GG06ER28" "IA10CO01"
[13] "IA10CO06" "KE06RD04" "LE05GI03" "LE05GI07" "LO08ZO13" "LU03IR19"
[19] "LU17UZ09" "be08rd25"
```

14 Distributions

```r
library(tidyr)
```

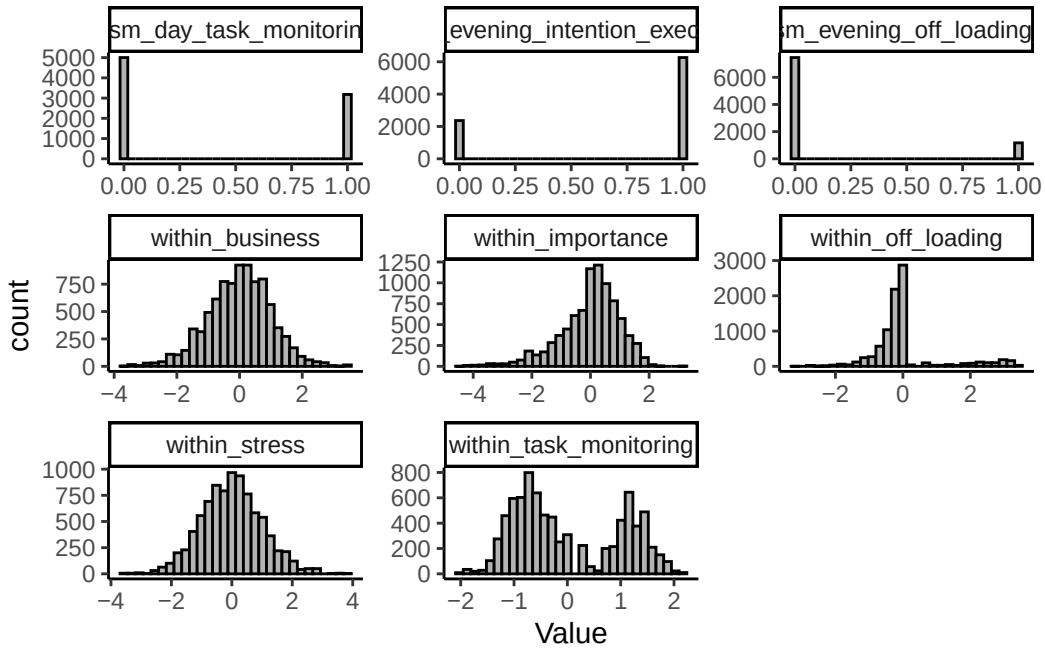
```
Attaching package: 'tidyr'

The following objects are masked from 'package:Matrix':

    expand, pack, unpack
```

```r
# Select the variables you want histograms for
df_selected <- stress_int_red %>%
  select(within_stress, within_importance, within_business,
         within_off_loading, within_task_monitoring,
         esm_evening_intention_execution,esm_day_task_monitoring,esm_evening_off_loading_r )
```

```
Adding missing grouping variables: `ID`, `day`, `intention`
```

```r
df_selected<-df_selected[, c("within_stress", "within_importance", "within_business",
                             "within_off_loading", "within_task_monitoring",
                             "esm_evening_intention_execution", "esm_day_task_monitoring",
                             "esm_evening_off_loading_r")]

# Convert to long format
df_long <- df_selected %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")

# Plot all histograms in one go using facets
ggplot(df_long, aes(x = Value)) +
  geom_histogram(bins = 30, fill = "grey70", color = "black") +
  facet_wrap(~ Variable, scales = "free") +
  theme_classic()
```

```
Warning: Removed 1962 rows containing non-finite outside the scale range
(`stat_bin()`).
```



15 Check the intraclass correlation coefficient (ICC)

```
PM_YA <- glmer(esm_evening_intention_execution ~ 1 + (1 | ID),
               data = stress_int[stress_int$agegroup=="YA",], family = binomial)

PM_MAA <- glmer(esm_evening_intention_execution ~ 1 + (1 | ID),
                data = stress_int[stress_int$agegroup=="MAA",], family = binomial)

print("ICC PM YA:")
```

```
[1] "ICC PM YA:"
```

```
icc_YA<-icc(PM_YA)
icc_YA$ICC_adjusted
```

```
[1] 0.1332206
```

```
print("ICC PM MAA:")
```

```
[1] "ICC PM MAA:"
```

```
icc_MAA<-icc(PM_MAA)
icc_MAA$ICC_adjusted
```

```
[1] 0.1889952
```

16 Test the effect of day on intention execution

17 Do people become more stressed as the go on with the study?

```
stress_time_model<-lmer(esm_day_stress_avg ~ day.c*agegroup.c + (day.c|ID),
                        data = stress_int_red,
                        control=lmerControl(optimizer="bobyqa",optCtrl=list(maxfun=100000)) )

tab_model(stress_time_model)
```

18 Conditional effects of variables on PM

Test the conditional effects of the variables on PM

```
# predict PM by all the predictors
PM_model<- glmer(esm_evening_intention_execution ~
                   # within factors
                   within_off_loading+
                   within_importance +
                   within_task_monitoring+
                   within_business+
                   within_stress+
                   # between factors
                   between_off_loading+
                   between_importance+
                   between_task_monitoring+
                   between_business+
```

```
                    between_stress+
                    # age
                    agegroup.c +
                    # day number as covariate
                    day.c+
                    # random effects
                    (1 within_off_loading+
                       within_importance +
                       within_task_monitoring+within_business+day.c| ID),
                  data = stress_int_red, family = binomial,
                  control=glmerControl(optimizer="bobyqa",optCtrl=list(maxfun=100000))
)
```

19 Results

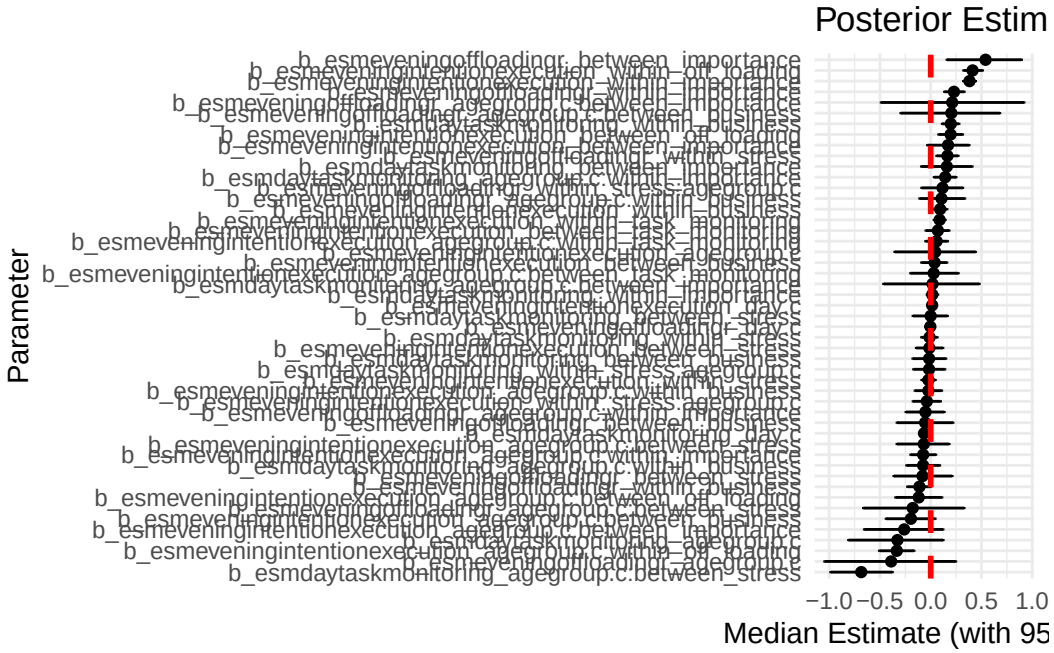
20 SEM - path model. Generalized linear-mixed effect path model - bayesian estimation

```
# first, we predict PM from all the predictors
PM_model <-bf(esm_evening_intention_execution ~  # fixed effects
                within_stress*agegroup.c +
                within_importance*agegroup.c +
                within_task_monitoring*agegroup.c +
                within_off_loading*agegroup.c +
                within_business*agegroup.c+
                between_stress*agegroup.c +
                between_task_monitoring*agegroup.c +
                between_off_loading*agegroup.c +
                between_importance*agegroup.c+
                between_business*agegroup.c+day.c+
                (1 + within_stress +
                   within_business+
                   within_importance +
                   within_task_monitoring +
                   within_off_loading || ID), # random intercepts and slopes
              family = bernoulli()) # we have a binary outcome

# predicting monitoring from stress and importance
monitoring_model<- bf(esm_day_task_monitoring ~
```

```
                          within_stress*agegroup.c +
                          within_importance*agegroup.c +
                          within_business*agegroup.c+
                          between_stress*agegroup.c+
                          between_importance*agegroup.c+
                          between_business+day.c+
                          (1+within_stress+
                             within_business+
                             within_importance+
                             day.c||ID),
                       family = bernoulli()) # mixture

# predicing off_loading from stress and imortance
offloading_model<-bf(esm_evening_off_loading_r ~
                        within_stress*agegroup.c +
                        within_importance*agegroup.c+
                        within_business*agegroup.c+
                        between_stress*agegroup.c+
                        between_importance*agegroup.c+
                        between_business*agegroup.c+ day.c+
                        (1+within_stress+
                           within_importance+
                           within_business+
                           day.c||ID), family = bernoulli() )#

# set weakly informative priors
priors <- c(
  # Fixed effects: assume standardized predictors
  prior(normal(0, 0.5), class = "b", resp = "esmeveningintentionexecution"),
  prior(normal(0, 0.5), class = "b", resp = "esmdaytaskmonitoring"),
  prior(normal(0, 0.5), class = "b", resp = "esmeveningoffloadingr")

# all the rest is left as defauls
 #
 #  # SD of random effects (hierarchical structure)
 #   prior( student_t(3, 0, 2.5) , class = "sd", resp = "esmeveningintentionexecution"),
 #  prior(student_t(3, 0, 2.5) , class = "sd", resp = "esmdaytaskmonitoring"),
 #   prior(student_t(3, 0, 2.5), class = "sd", resp = "esmeveningoffloadingr"),
 #  #
 #  # Correlation priors among random effects
 # # prior(lkj(2), class = "cor"),  # same across responses
 #
```

```
 #  # Residual SDs for continuous outcomes
 #  prior(exponential(1), class = "sigma", resp = "esmdaytaskmonitoring"),
 #  prior(exponential(1), class = "sigma", resp = "esmeveningoffloadingr")
)


# fit thestress_int_red# fit the model
fit1 <- brm(
  PM_model+
    monitoring_model +
    offloading_model +
    set_rescor(F),
  prior = priors,
  sample_prior = "yes",
  data = stress_int_red,
  chains = 4, cores = 4,
  iter = 2000,
  warmup = 1000
  #,backend = "cmdstanr"
)
```

21 Posterior summary




```
Warning: package 'ggdist' was built under R version 4.4.3


Attaching package: 'ggdist'

The following object is masked from 'package:bayestestR':

    hdi

The following objects are masked from 'package:brms':

    dstudent_t, pstudent_t, qstudent_t, rstudent_t
```

22 Path model results

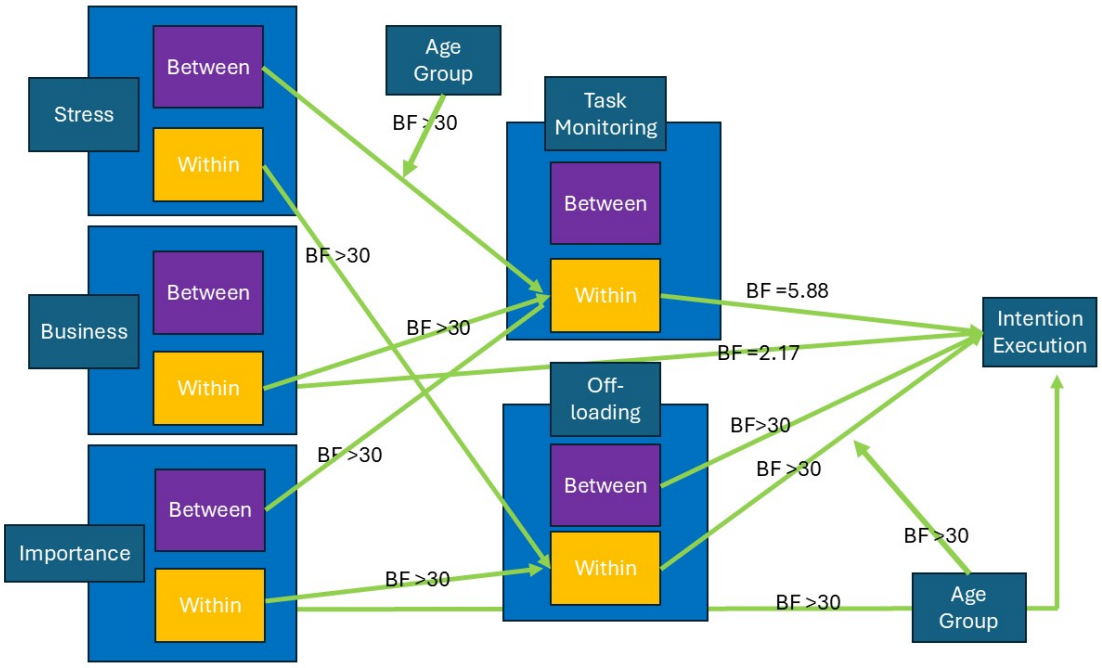


Figure 7: Extended hypothesised path model

23 Moderation of Age group on between stress-within task monitoring path

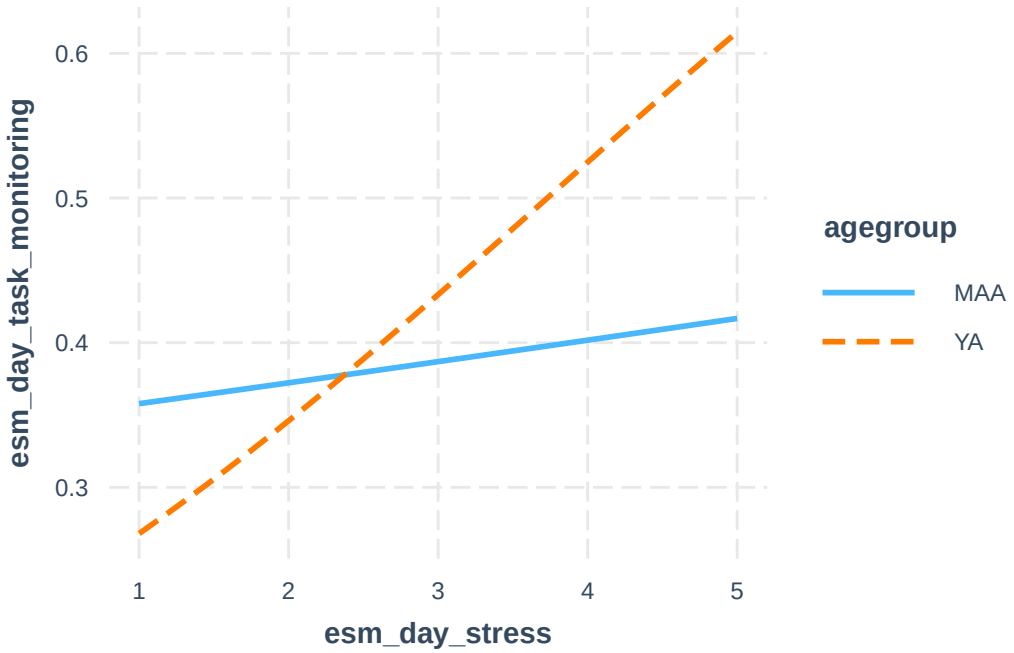
```
int_mod_stess_mon<-glmer(esm_day_task_monitoring~between_stress*agegroup +(1|ID),
               family = binomial(),
               data = stress_int_red, )


tab_model(int_mod_stess_mon)
```

```
int_mod_stress_mon<-glmer(esm_day_task_monitoring~day.c+ esm_day_stress*agegroup +(1+day.c+es
               family = binomial(),
               data = stress_int_red )
```

```
boundary (singular) fit: see help('isSingular')
```

```r
tab_model(int_mod_stress_mon)
```

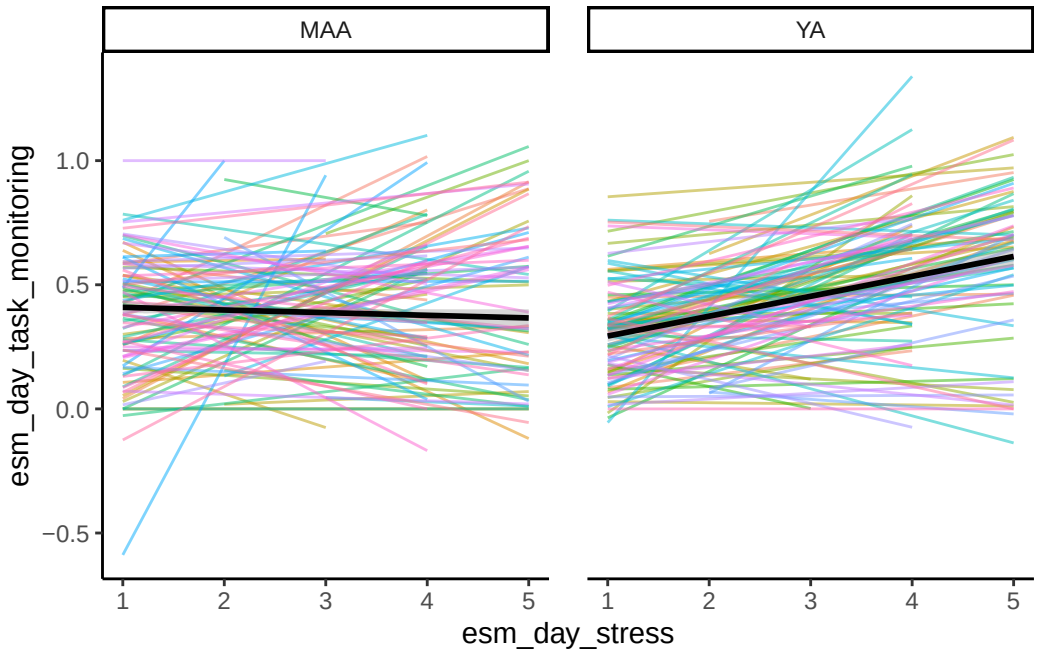
```r
interact_plot(int_mod_stress_mon, pred = esm_day_stress, modx = agegroup)
```




```r
ggplot(stress_int_red,
        aes( x=esm_day_stress, y=esm_day_task_monitoring))+
 # add the "smooth" line, which the regression method ('l,')
 # and trasparent (0.5)
   geom_line(stat="smooth",method = "glm", formula=y~x, alpha=0.5, se=F)+

 # specify that we want different colours for different participants
   aes(colour = factor(ID))+
 # add the summary line with geom_smooth
   geom_smooth(method="lm",formula=y~x, se=T, colour = "black" )+
   theme(strip.text.x = element_text(size = 13))+
   theme_classic()+
   theme(panel.spacing = unit(1, "lines"))+
   facet_wrap(.~agegroup)+
   #ggtitle("Experiment 2")+
   theme(legend.position = "none")
```

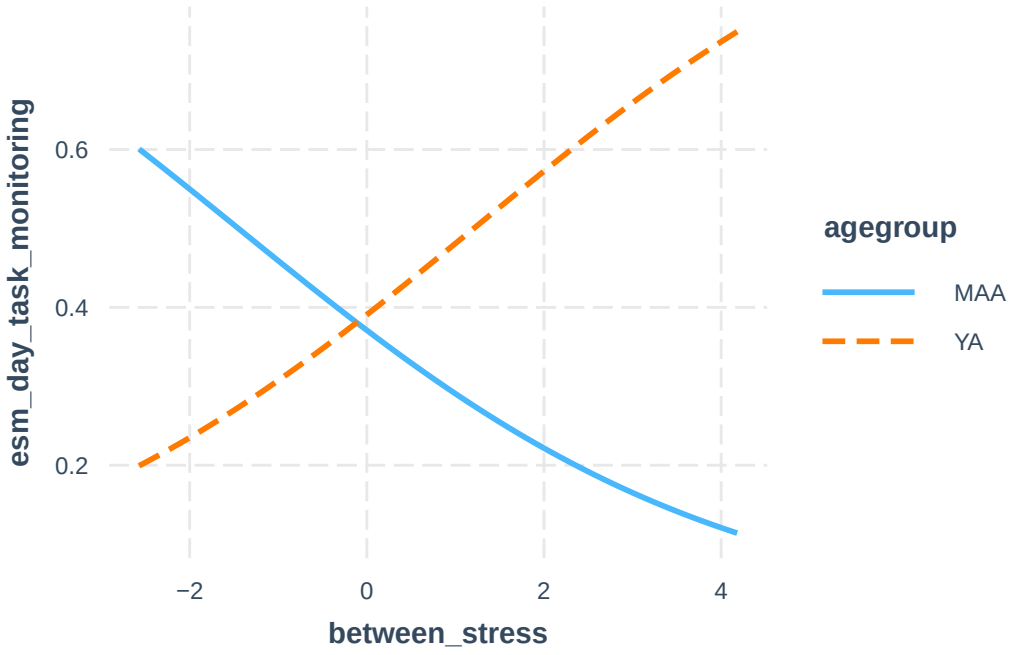
```
Warning: Removed 635 rows containing non-finite outside the scale range
```

```
(`stat_smooth()`).
Removed 635 rows containing non-finite outside the scale range
(`stat_smooth()`).
```



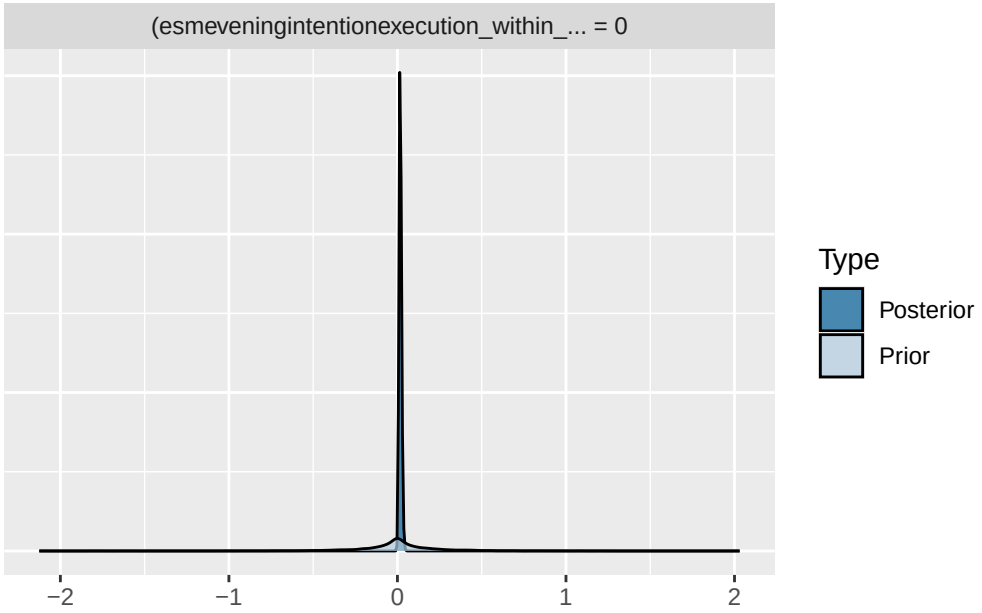
24 Plot moderation of Age group on between stress-within task monitoring path

```
library(interactions)
interact_plot(int_mod_stess_mon, pred = between_stress, modx = agegroup)
```



25 Test the mediation of monitoring

```
h_business_monitoring<-hypothesis(fit1,  c("esmeveningintentionexecution_within_task_monitori

plot(h_business_monitoring)
```




```
print(h_business_monitoring)
```

```
Hypothesis Tests for class b:
                Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio
1 (esmeveningintent... = 0     0.02      0.01     0.01     0.03       0.18
  Post.Prob Star
1      0.15    *
---
'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
'*': For one-sided hypotheses, the posterior probability exceeds 95%;
for two-sided hypotheses, the value tested against lies outside the 95%-CI.
Posterior probabilities of point hypotheses assume equal prior probabilities.
```

```
print(paste0("Bayes Factor for ", "h_mediation"," is ", 1/h_business_monitoring$hypothesis$Ev
```

```
[1] "Bayes Factor for h_mediation is 5.56545497639755"
```

26 Test moderated mediation - Age as moderator, off-loading as a mediator

```
h_moderated_mediation<-hypothesis(fit1,  c("esmeveningintentionexecution_agegroup.c:within_o

BF_moderated_med<-1/h_moderated_mediation$hypothesis$Evid.Ratio
print(paste0("Bayes Factor for ", "h_moderated_mediation"," is ", 1/h_moderated_mediation$hyp
```

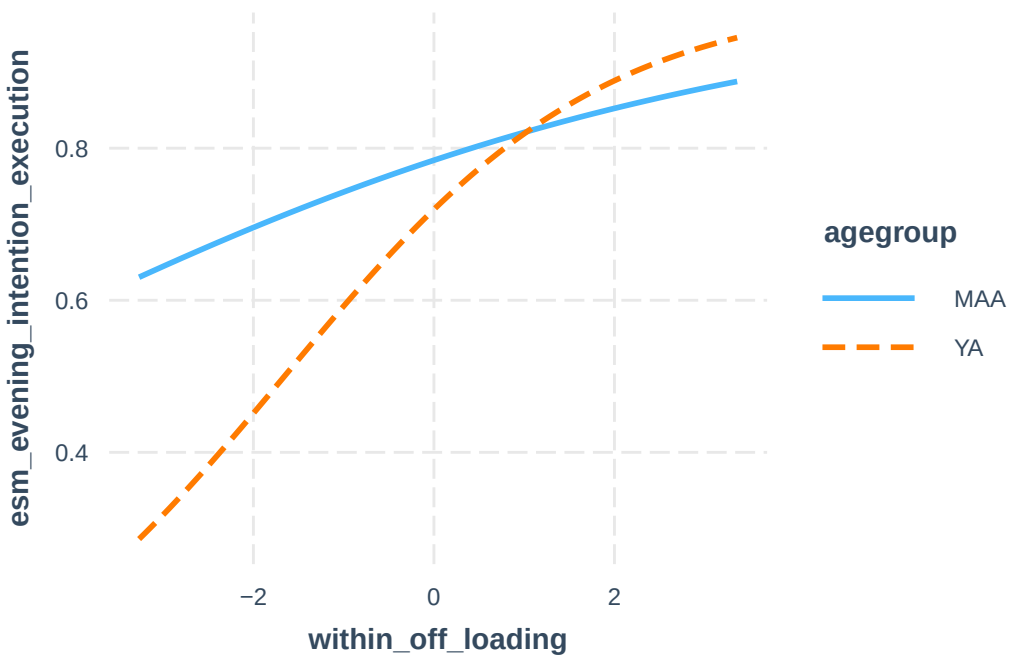
27 Moderation model

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: esm_evening_intention_execution ~ within_off_loading * agegroup +
    (within_off_loading | ID)
   Data: stress_int_red

     AIC      BIC   logLik deviance df.resid
  9571.4   9620.9  -4778.7   9557.4     8619

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.6160 -0.9707  0.4621  0.6220  2.0074

Random effects:
 Groups Name               Variance Std.Dev. Corr
 ID     (Intercept)        0.43321  0.6582
        within_off_loading 0.06023  0.2454   -0.42
Number of obs: 8626, groups:  ID, 210

Fixed effects:
                              Estimate Std. Error z value Pr(>|z|)
(Intercept)                    1.29017    0.07643  16.880  < 2e-16 ***
within_off_loading             0.23164    0.05461   4.242 2.22e-05 ***
agegroupYA                    -0.34823    0.10549  -3.301 0.000963 ***
within_off_loading:agegroupYA  0.33695    0.08180   4.119 3.80e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
            (Intr) wthn__ aggrYA
wthn_ff_ldn -0.084
agegroupYA  -0.714  0.081
wthn_ff_:YA  0.080 -0.606 -0.068
```

28 Plot moderation

```
interact_plot(int_mod, pred = within_off_loading, modx = agegroup)
```




```
# # get the posterior samples
# draws<-posterior_samples(fit1)
#
# # Compute indirect effect manually
# draws$indirect <- draws$b_withinoffloading_within_importance * # path a - M~A
#   draws$b_esmeveningintentionexecution_within_off_loading  # path b - Y~M

# # plot distribution
# ggplot(draws, aes(x = indirect))+
#   geom_histogram()+
```

```
#   theme_classic()+
#   gg_title("posterior distribution indirect effect of intention on PM")

#indirect_bf <- bayesfactor_parameters(draws$indirect)

#plot(bayesfactor_parameters(draws$indirect))

# draws$indirect2 <- draws$b_withintaskmonitoring_within_business*  draws$b_esmeveningintenti
#
#
# hist(draws$indirect)
#
# hist(draws$indirect2)
#
# # simple slope analysis for the interaction between age group and within task monitoring
# slope_MA <- draws$b_esmeveningintentionexecution_within_off_loading  +
#   draws$`b_esmeveningintentionexecution_agegroup.c:within_off_loading` * 0.5
# slope_YA <- draws$b_esmeveningintentionexecution_within_off_loading  +
#   draws$`b_esmeveningintentionexecution_agegroup.c:within_off_loading`  -0.5+0.5
#
# describe_posterior(slope_MA)
# describe_posterior(slope_YA)
#
# conditional_effects(fit1, "within_off_loading", resp = "esmeveningintentionexecution")
#
# ce <- conditional_effects(fit1, effects = "within_off_loading:agegroup.c",
#                           conditions = data.frame(agegroup.c = c(-0., 0.52)))
#
# plot(ce)

# stronger in young adults than in middle aged adults
```

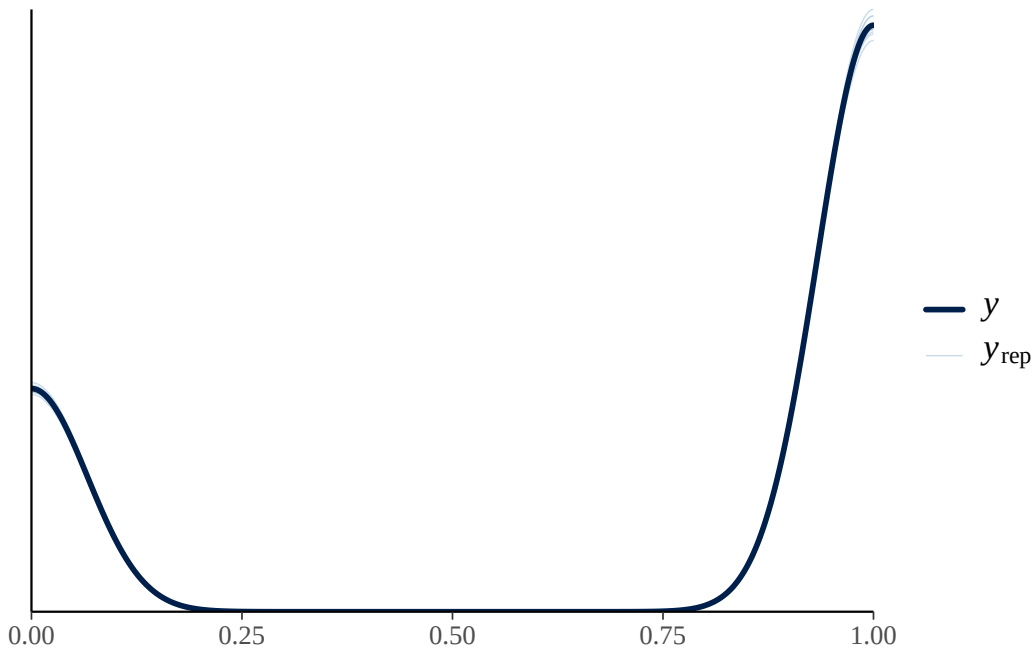
29 Posterior predictive check

PPC constructs the posterior distribution over the parameters and simulate their distribution - if it approximates the distribution of observed data, the model is a good fit

Prospective memory

```
pp_check(fit1, resp = "esmeveningintentionexecution")
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

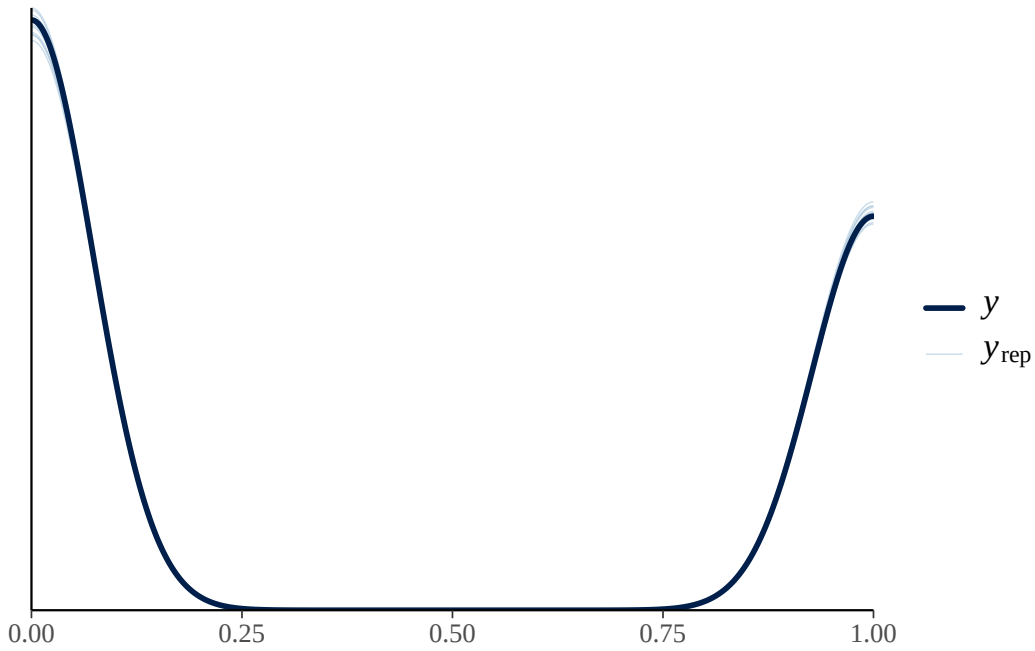


30 Posterior predictive check

Task Monitoring

```
pp_check(fit1, resp = "esmdaytaskmonitoring")
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

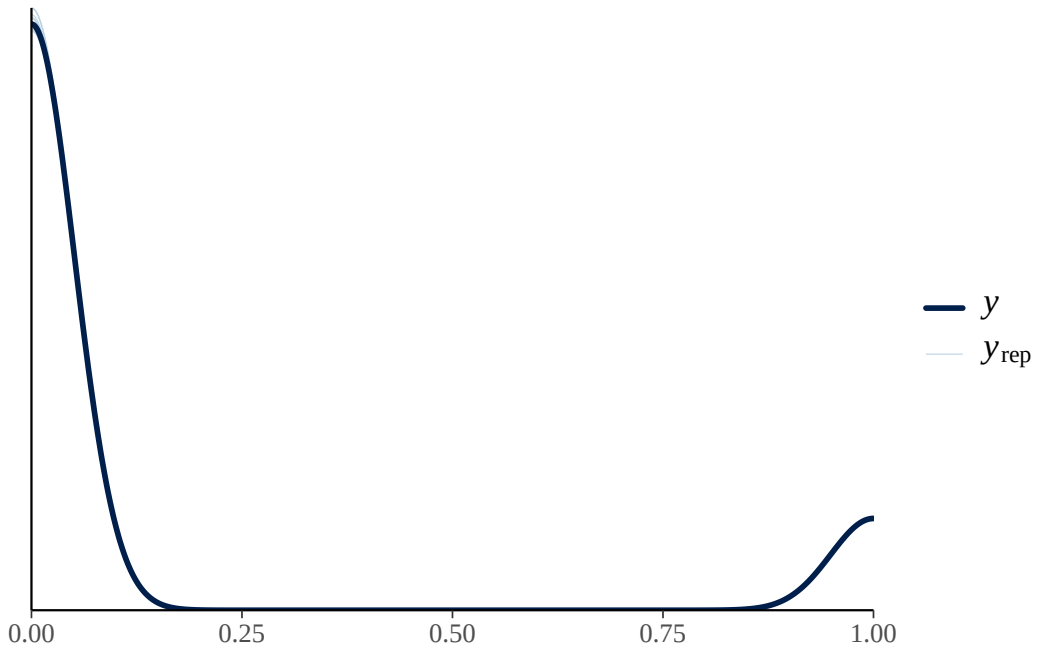


31 Posterior predictive check

Off Loading

```
pp_check(fit1, resp = "esmeveningoffloadingr")
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```


0.00
0.25
0.50
0.75
1.00
y
y_{rep}